

EDUCATION

Heritage High School, Frisco, TX

Class of 2027

GPA 4.00 UW / 5.361 W · Rank 11/481 · ACT 36 · AP Scholar with Distinction

AP exams: 9 taken, a 5 on all 9 — Computer Science A, Statistics, Physics 1, Seminar, English Language & Composition, U.S. History, World History, Psychology, Human Geography

Coursework: AP Calculus BC, AP Physics 2, AP Physics 1, AP Computer Science A, AP English Language, AP English Literature, AP Government, AP Economics, Computer Science III, Mobile App Programming, Spanish 2

Competitions: Semifinalist, Diamond Challenge 2026 (X-Step) · Finalist, [Texas High School Ideas Challenge](#) at Texas A&M (X-Step, May 2026) · 1st, TechLit Bridging the Gap Hackathon ([ShopSmart](#), Aug. 2025) · Innovation Prize, PeddieHacks (235 participants) · 3rd, Hack4Humanity deeptanshudevatha.com/projects/competitions

RESEARCH

PAPER 1 · COMPUTATIONAL LINGUISTICS / CLINICAL ML

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features[↗] May 2025–Present

First Author · Mentor: Joe Xiao (PhD student)

medRxiv preprint (Aug. 2026) medrxiv.org/content/10.64898/2026.08.04.26359730v1 · Under peer review, *Frontiers in Aging Neuroscience* — submitted August 04, 2026 · Public code github.com/ddevatha1/dementia-mmse-prediction

QUESTION	Can transcribed speech plus the clinical history already on file forecast a patient's next-visit MMSE (0–30)? Framed as longitudinal, patient-level supervised regression; most prior speech work is single-visit classification.
DATA	DementiaBank Pitt Corpus (controlled access): 126 patients (69 dementia, 57 control), 2–5 visits each → 173 prediction samples.
FEATURES	Designed a 53-dimensional feature vector: 12 per-visit features — Sentence-BERT discourse coherence (adjacent-sentence similarity, global coherence, semantic slope, topic drift) plus spaCy lexical/syntactic measures (filler rate, 50-token MATTR, inter-sentence repetition, POS-based fragment ratio, named-entity rate) — with 8 visit-to-visit Δ features, 4 temporal features, 20 PCA components of 384-d SBERT embeddings, and 9 clinical/demographic covariates.
MODEL	LightGBM, chosen for small mixed-type tabular data: RobustScaler → correlation pruning ($ r > 0.8$, clinical variables kept) → permutation-importance selection (top 15) → 3× Gaussian augmentation tagged by parent patient → Optuna TPE over 10 hyperparameters, 200 trials, maximizing mean patient-grouped R^2 .
VALIDATION	<u>Diagnosed patient-identity leakage</u> : folds split by visit, so a model could score by recognizing a person. Rebuilt evaluation as 5-fold GroupKFold on patient ID, locking augmented copies to their parent's fold and scoring only out-of-fold predictions on original samples; scores fell once leaked identity was removed.
DEBUGGING	Found why entry MMSE, the expected top predictor, contributed almost nothing: <u>two silent data-join bugs</u> (renamed column; string-vs-integer patient ID) imputed one value for every patient. Fixing them was the largest single gain.
RESULTS	R^2 0.840 ± 0.015 · RMSE 2.75 · MAE 2.02 MMSE points · Pearson r 0.917. Entry MMSE, baseline diagnosis and Blessed score lead permutation importance.
ABLATION	Controlled feature-source ablation: linguistic-only R^2 0.118, SBERT-only 0.458, clinical-only 0.716, clinical + SBERT 0.759. Clinical history dominates; SBERT beats handcrafted features and adds complementary signal.
INTEGRITY & LIMITS	Audited the manuscript against the code, removing six unused features. Limits: non-nested selection and tuning (CV may be optimistic); one cohort, no external validation; over-prediction below MMSE 10.

ACCEPTED FOR PRESENTATION — IEEE-EMBS IECBES 2026 KUALA LUMPUR · DEC. 9–11, 2026

A Four-Site Plantar-Pressure Insole: Transducer Error Budget, Feature Equivariance, and Sparse-Versus-Dense Agreement in Overground Walking[↗] Jul. 2026–Present2nd of 5 authors · ML lead · MIT CSAIL first author; [UC Santa Cruz AIEA Lab](#) (Prof. Leilani Gilpin) · deeptanshudevatha.com/research/x-step

FRAMING	Asks what the team's existing four-site insole (MET1, MET2, MET5, heel; ESP32, 25 Hz) can legitimately measure — a measurement instrument, not a clinical classifier. I did not design the hardware; my work starts from the pressure data it produces. No wearer data yet; no clinical claim.
ML PIPELINE	From those readings I built the analysis pipeline: the feature map over rolling 4 s windows (100 samples × 8 channels) to 59 features — per-site peak, mean, coefficient of variation and pressure-time integral; window-level cadence, stance ratio, anterior load share, aggregate peak and total PTI — and a 540-parameter logistic-regression head (≈ 0.23 ms per update).
CORRELATION STUDY	Designed the sparse-vs-dense study against an independent 32-cell insole archive (149 recordings, 15 adults, 298 foot-takes), comparing the cell nearest each of the insole's four sites with its anatomical cluster. Shape survived (median r 0.72–0.99), magnitude did not (peak-to-peak r 0.43–0.74); four-site center of pressure tracked anteroposterior (r 0.81) but not mediolateral motion (r 0.01) — a limit of the existing layout's geometry. Descriptive only.
EVALUATION	Quantified subject-identity leakage on a synthetic 24-subject, 9-class cohort: macro-F1 0.885 subject-grouped vs 0.931 ungrouped (majority baseline 0.040) — a gap larger than the spread across classifiers.

PROJECTS & LEADERSHIP

ShopAI [↗] — *Creator & Sole Developer* · live on the [Apple App Store](#) 2025–Present
One grocery search across nine retailers, plus multi-store trip planning · solo rebuild of a team hackathon prototype · [Architecture write-up](#)

ARCHITECTURE	React Native/Expo client over a standalone Express/TypeScript backend on AWS EC2 behind Nginx ; the client holds no search logic — every search, plan and warm-up check runs server-side through a four-layer, one-direction pipeline.
ADAPTERS	Designed a retailer-adapter abstraction normalizing nine chains (seven with live data) into one schema via four integration strategies: OAuth2 client-credentials REST (Kroger, Harris Teeter), anonymous-session GraphQL (Aldi, Sprouts, Publix via Instacart), headless Playwright (Trader Joe's), server-rendered HTML (Whole Foods). Target excluded (terms of service).
SEARCH PATH	Query correction → concurrent fan-out racing a shared response-time budget → per-store food/relevance filtering and ranking → explicit ok / error / unavailable / pending status per store. A per-ZIP state machine (starting → warming → ready) pre-warms OAuth tokens, cookies and the browser session off the request path.
PLANNER	A deterministic optimizer exhaustively scores all $2^7 - 1 = 127$ non-empty store subsets, so “cheapest” is exact; cheapest, fastest and fewest-stops are single-field sorts; only “balanced” is weighted multi-criteria. Zero LLM calls in search or planning.
PRODUCTION	Debugged a Node.js setTimeout overflow in a 30-day location cache that stalled the event loop (≈ 1.5 s → 27 s searches) by capping the sweep interval; shipped by feature-flagging off unfinished meal planning, AI assistant, OCR and barcode scanning.

KidCode [↗] — *Founder & Lead Instructor* · Frisco, TX Jun. 2024–Present
124 students · 3-year [program](#) [deeptanshudevatha.com/projects/kidcode](#) · 10 Frisco neighborhoods · **120-class-day curriculum** · **20 projects** · median 10 per student (max 18)

CURRICULUM	Designed a three-year, 120-class-day, 20-project (6 + 7 + 7) curriculum: Year 1 Python foundations; Year 2 assert-based unit tests, file I/O, JSON, exception handling, classes and inheritance, libraries; Year 3 spaCy NLP — tokenization, lemmatization, sentence segmentation, POS tagging, NER, Matcher rule patterns, and a rule-based chatbot.
SPIRAL DESIGN	Earlier projects return at a higher level: Mad Libs becomes Auto Mad Libs with spaCy choosing the blanks; the trivia quiz returns JSON-backed with error handling; Year 2 ends by breaking students' own text-cleaning code, which Year 3 opens by testing against spaCy.
ASSESSMENT	Wrote <u>13 curriculum-aligned pre/post assessment pairs</u> (same skills and formats, different code). Paired results, n = 104 : mean 41.7% → 58.8% (+17.1 pp) ; 99/104 improved, 4 unchanged, 1 declined. Observed change only; no comparison group.
STUDENT WORK	Students completed 1,214 working programs across the <u>20 projects</u> , from Python basics in Year 1 to spaCy NLP in Year 3 (all 572 Year 3 programs use spaCy).

X-STEP STARTUP · PRODUCT BUILT ON THE PAPER 2 INSOLE

X-Step [↗] — *Co-Founder & ML/Mobile Lead* · student startup team · working prototype, no wearer data yet Aug. 2025–Present
[deeptanshudevatha.com/projects/x-step](#)

PRODUCT	A plantar-pressure insole for people at risk of diabetic foot problems, built by the team: four FSR402 sensors per foot on an ESP32 at 25 Hz, streaming sequence-numbered 28-byte BLE frames to a React Native (Expo) app with live pressure, trends, alerts and a clinician view.
MY ROLE	Wrote the React Native/Expo app (BLE client, live pressure heatmap, alerts, trends, guided routine; earlier Flask + LLM health-assistant backend) and trained two CNN prototypes on public data (heatmap postures, wound severity). Designed the ML pipeline that turns the stream into alerts: rolling 4 s windows → 59 pressure and gait features → 540-parameter logistic-regression head, fast enough to run between 40 ms samples.

Ayusha Nilayam [↗] — *Programming & Math Instructor* · Hyderabad, India (remote) Nov. 2025–Jul. 2026
[deeptanshudevatha.com/projects/ayusha-nilayam](#)

SCOPE	Taught <u>38 weekly Zoom sessions</u> of Python and math to 11 children at a children's home, plus two weeks in person.
CONTENT	Python from print() through conditionals, loops and functions to independent projects (calculator, number-guessing game); math from equations and inequalities through the unit circle, and statistics through standard deviation.
CONSTRAINTS	Adapted to one shared laptop, unreliable internet and a Telugu–English language gap with frequent checks for understanding and student explain-backs.

UT Dallas NLP with Python Workshop — *Participant* Summer 2025
Instructor: Dr. Chris Davis · Built a rule-based text simplifier: dependency parsing, lexical substitution, clause splitting, passive-to-active rewriting.

TECHNICAL SKILLS

ML & NLP	LightGBM, scikit-learn, Optuna (TPE), PyTorch, PCA, permutation importance, GroupKFold, ablation design; spaCy (tokenization, POS, NER, Matcher, dependency parsing), Sentence-BERT; windowed sensor-feature extraction
LANGUAGES	Python, TypeScript/JavaScript, Java, SQL
SYSTEMS	React Native/Expo, Node.js/Express, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx